

The Hill sphere transits of β Pictoris c and the search for another 1981-like event

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ABSTRACT

Context. β Pictoris is a young and nearby planetary system hosting an edge-on debris disk and at least three gas-giant planets. Their circumplanetary environments could host exomoons and rings, a detection of which would be highly informative for moon and planet formation theories. A photometric fluctuation of about 4% was seen towards β Pictoris in 1981, indicating the transit of dust in the system, which could be associated with the Hill spheres of the two inner planets.

Aims. We search for the origin of the 1981 event by searching for an analogous event in multi-epoch photometry from 2017 to 2023, and look for signs of circumplanetary material in transits of the Hill sphere of β Pictoris c.

Methods. Observations from the BRITE satellite, and the bRing and ASTEP observatories are fitted to a model of the 1981 event to search for a similar event, and also search for a signal consistent with a circumplanetary disk transit using a simple flat disk model.

Results. No compelling evidence for a 1981 event during a Hill sphere primary transit is found, although we do find a candidate event around 2019 July 26. Due to the uncertainty in the time of closest projected separation of β Pictoris c, a search for a disk during the primary transit of 2018 could not be robustly determined, but bRing photometry for the second primary transit places an upper limit on the dust content of the Hill sphere of $\sim 10^{21}$ grams of material.

Conclusions. With no compelling detection of an event similar to that seen in 1981 during the transits of β Pictoris c in the photometric time series, we rule out the hypothesis that this event was related to the Hill spheres of β Pictoris c and β Pictoris b. Future observations of the next Hill sphere transit in 2028 can be realised with both ground-based observatories and with PLATO, whose first long duration science pointing will include β Pictoris.

Key words. planets and satellites: rings – planets and satellites: formation – stars: individual: β Pictoris

1. Introduction

A circumplanetary disk (CPD) forms around a planet (?) within its Hill sphere, defined as the radius where matter is gravitation-

ally bound to the planet. In addition to dust and gas within the CPD being accreted onto the planets, it can dissipate by radiation pressure or the dust can collide and fragment (?). The collisions can lead to dust clumping and eventually the creation of

exosatellites, which continue to accrete material, creating gaps and rings. In this paper we focus on two Hill sphere transits of β Pictoris c.

β Pictoris (HD 39060, HIP 27321) is a bright ($V=3.85$), nearby ($d=19.6$ pc), young (~ 23 Myr-old) star with an exoplanetary system of two known planets and a complex circumstellar disk (see Table 1; ESA 1997) (????). After the discovery of its edge-on debris disk (?), it is one of the first disks to be spatially resolved, dominating the research field of debris disks (?). One of the first major discoveries is the inner warped disk at 5° from the main disk (?). The primordial dust from the interstellar molecular cloud has likely been dispersed, leaving a system composed of disintegrating bodies in an early-phase planetary system (e.g. ????). Moreover, star-grazing comets have provided evidence of their collisions and evaporation, giving rise to a mix of gas and dust in the disk (e.g. ?????, and references therein).

The structure and composition of the disk led to the belief that the system hosts exoplanets, and in 2009 the first exoplanet in the system was detected by means of high-contrast imaging (?). The exoplanet β Pictoris b was found to have a mass of $11 M_{\text{Jup}}$ and a period of 21 years orbiting at 9.8 au (e.g. ?????, and many more). Several years later β Pictoris c was discovered through the radial velocity technique (???), where ? found the exoplanet to be of comparable size to β Pictoris b with a mass of $7.8 \pm 0.4 M_{\text{Jup}}$ orbiting at closer proximity to the star with a semi-major axis of 2.7 ± 0.02 au. Astrometry of β Pictoris b was used to derive the astrometry of β Pictoris c, giving an approximate orbital period of 3.3 years (?). Most recently, a third gas giant planet, β Pictoris d has been simultaneously detected by two groups (??) on a wide ~ 25 au orbit with a mass in the $2\text{--}4 M_{\text{Jup}}$ range.